HUMAIRA ANZUM NEHA

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Technical Summary

- Research experience in human-centered AI and accessible technology emphasizing on people with visual impairments and people who stutter.
- Research experience in qualitative HCI analysis, conducting semi-structured interviews and thematic analysis.
- Research experience using deep learning models (e.g., YOLOv5, ResNet-50) for object detection, currency recognition, and face recognition in computer vision.
- Research experience in assistive wearables integrating the Raspberry Pi Compute Module 4 (CM4) with distance and motion sensors to support obstacle awareness and mobility assistance.
- Developed a mobile application integrated with a deep learning model for non-invasive glucose and hemoglobin estimation through fingertip video processing.
- Developed a training free method to improve visual prompting for long-video understanding across different VLMs.
- Developed a multimodal fusion pipeline for audio and text using graph neural networks and LLM.
- Hands-on experience in custom image dataset development that includes data collection, annotation, and preparation.
- Hands-on experience on deploying deep learning models on embedded platforms for on-device processing.
- Proficient in building and working with deep learning models using TensorFlow, Keras, PyTorch, and Hugging Face.

Education

M.Sc in CSE at BRAC University **Jan. 2026 – June 2027 (Exp.)**
Thesis: Speech Emotion Recognition for Atypical Speech *Dhaka, Bangladesh*

- CGPA: 4.00/4.00, up to 2nd semester.

B.Sc. in CSE at KUET **Feb. 2019 – Mar. 2024**
Thesis: Real-Time Object, Currency, and Face Recognition for Blind and Low-Vision People *Khulna, Bangladesh*

- CGPA: 3.87/4.00, ranked 3rd in the department.

Skills

Programming Languages: Python, C, C++, Java, Kotlin, Javascript, Bash, SQL.
Deep Learning Libraries: TensorFlow, Keras, PyTorch, Hugging Face
Computer Vision: OpenCV, TorchVision, ONNX Runtime
AI Tools: OpenAI Codex, Claude.
Developer Tools: Anaconda, Jupyter, Arduino, Raspberry Pi , Git, MS Visual C++, VS Code.
Database: MySQL, Oracle, Firebase, MongoDB
Mobile Apps Development: Android Studio, CameraX
Operating System: Linux, MacOs, Windows.
Research Aptitude: Human-Centered AI, Accessibility, Computer Vision, and Multimodal AI
Adaptive Skills: Academic Writing, Collaborative Research, Community-Engaged Research, Complex Problem Solving, Mentoring and Leadership, Rapid Skill Acquisition.

Experience

Lecturer **Aug. 2024 – Present**
Computer Science and Engineering, United International University *Dhaka, Bangladesh*

- Led a project evaluating how reliably vision-language models read and interpret expiration dates in diverse formats for blind and low-vision users, and exploring ways to improve model performance.
- Oversaw undergraduate courses including Data Structures and Algorithms, Database Management Systems, Data Mining, Object Oriented Programming, Digital Image Processing, and Computer Architecture.
- Supervised undergraduate thesis projects, mentored competition teams, judged departmental project showcases, and supported student learning through coursework evaluation, feedback, and office hours.

Projects

Query-Guided Visual Prompting for Long Video Understanding | *Python, PyTorch, VLMs* **Feb. 2026 - Present**

- Designed a frame scoring method using a lightweight vision-language model (CLIP) to select question-relevant frames and arrange them into asymmetric grids that give more space to more relevant frames and used them as visual prompts for vision-language models without any fine-tuning.
- Improved pair accuracy on the MF2 benchmark from 16.9% to 23.5% over the baseline using Qwen2.5-VL-7B.

Video Creation and Editing among People Who Stutter | *Semi-Structured Interview* **May 2026 - Sep. 2026**

- Conducted an onboarding workshop and semi-structured interviews with eight people who stutter in Bangladesh to understand how they engage in video content creation to represent themselves and their community.
- Analyzed the interview data using reflexive thematic analysis to uncover how fluency norms, video context and audience influence editing choices and what support they need from AI-assisted editing tools.

GNN–BERT Fusion for Understanding Music Context | *PyTorch, GNN, BERT* **Jan. 2026 - April 2026**

- Leveraged graph neural network models to build an audio graph and fused them into a pre-trained large language model (BERT) for textual feature extraction and class tagging.
- Conducted ablation studies to assess the impact of graph-based representations and audio-text fusion methods.
- Achieved a 21.8% relative improvement in AUC-PR using cross-attention fusion over the best single-modality baseline on MagnaTagATune dataset.

Assistive Vision Enabling Object, Currency, and Face Recognition | *PyTorch* **July 2023 - June 2024**

- Trained real-time object detection models (YOLOv5) on a custom annotated dataset to detect objects and classify Bangladeshi banknotes and deployed the trained model on Raspberry Pi.
- Leveraged ResNet-50 embeddings and cosine similarity for face recognition and improved face detection in low-light conditions by applying histogram equalization.
- Achieved 98.2% mAP for objects, 99.7% top-1 accuracy for currency, and 97.6% accuracy for face recognition.

Assistive Glasses for Navigation, Mobility, and Reading | *Raspberry Pi, Pytesseract* **June 2023 - March 2024**

- Integrated distance sensors (Time-of-Flight) with a motion sensor (accelerometer) on Raspberry Pi to support obstacle awareness and mobility assistance, providing real-time alerts to users and caregivers.
- Implemented an optical character recognition (OCR) and text-to-speech pipeline that reads printed text aloud to users with 93.0% character accuracy.

Mobile Application to Measure Blood Component Levels | *Kotlin, Python, OpenCV* **Dec. 2022 - May 2023**

- Developed a mobile application that used a deep learning model to analyze a 15-second fingertip video recording and extracts photoplethysmogram (PPG) signal to estimate glucose and hemoglobin levels using a deep learning model.
- Calibrated camera configurations, including frame rate, exposure, and illumination settings, for consistent video quality across lighting conditions and devices.

Abalone Board Game with AI Opponent | *Python, Pygame* **Nov. 2022 - Feb. 2023**

- Developed a board game to play with an AI opponent using Minimax search and alpha–beta pruning.
- Designed a heuristic evaluation function to assess board states by considering the difference in marble counts between players, assigning higher weights to central positions, and rewarding moves that push opponent marbles off the board.

Relevant Coursework

- Natural Language Processing
- Deep Learning for Computer Vision
- Neural Networks and Fuzzy Systems
- Research Methodology (HCI-Focused)
- Fairness, Ethics and Explainability in AI
- Computer and Network Security
- Data Mining
- Mobile Computing
- Artificial Intelligence
- Embedded Systems
- Image Processing
- Applied Statistics

Publications

- **H. Neha**, M. S. Sadi, M. R. Islam and S. Sakib, "Automated Object Identification for the Visually Impaired," 2024 27th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 20-22 Dec., 2024, pp. 3584-3589, doi: 10.1109/ICCIT64611.2024.11022362.
- **H. Neha**, S. Sakib, F. Sadaf and S. M. Taslim Uddin Raju, "Mobile Application to Collect Data and Measure Blood Component Level in a Non-Invasive Way," 2023 26th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 13-15 Dec., 2023, pp. 1-6, doi: 10.1109/ICCIT60459.2023.10441186.
- S. Sakib, M. S. Sadi, M. S. Arefin, M. R. Islam and **H. Neha**, "An Assistive Device for the Visually Impaired for Enhanced Navigation and Reading," 2024 13th International Conference on Electrical and Computer Engineering (ICECE), Dhaka, Bangladesh, 18-20 Dec., 2024, pp. 767-772, doi: 10.1109/ICECE64886.2024.11024586.

Achievements

- Dean's Award for excellent academic performance in KUET for academic years 2019–20, 2020–21, and 2021–22.
- Selected as a Top 10 finalist in the 2021 UNDP women's digital innovation hackathon for developing a community application for parents of autistic children.